

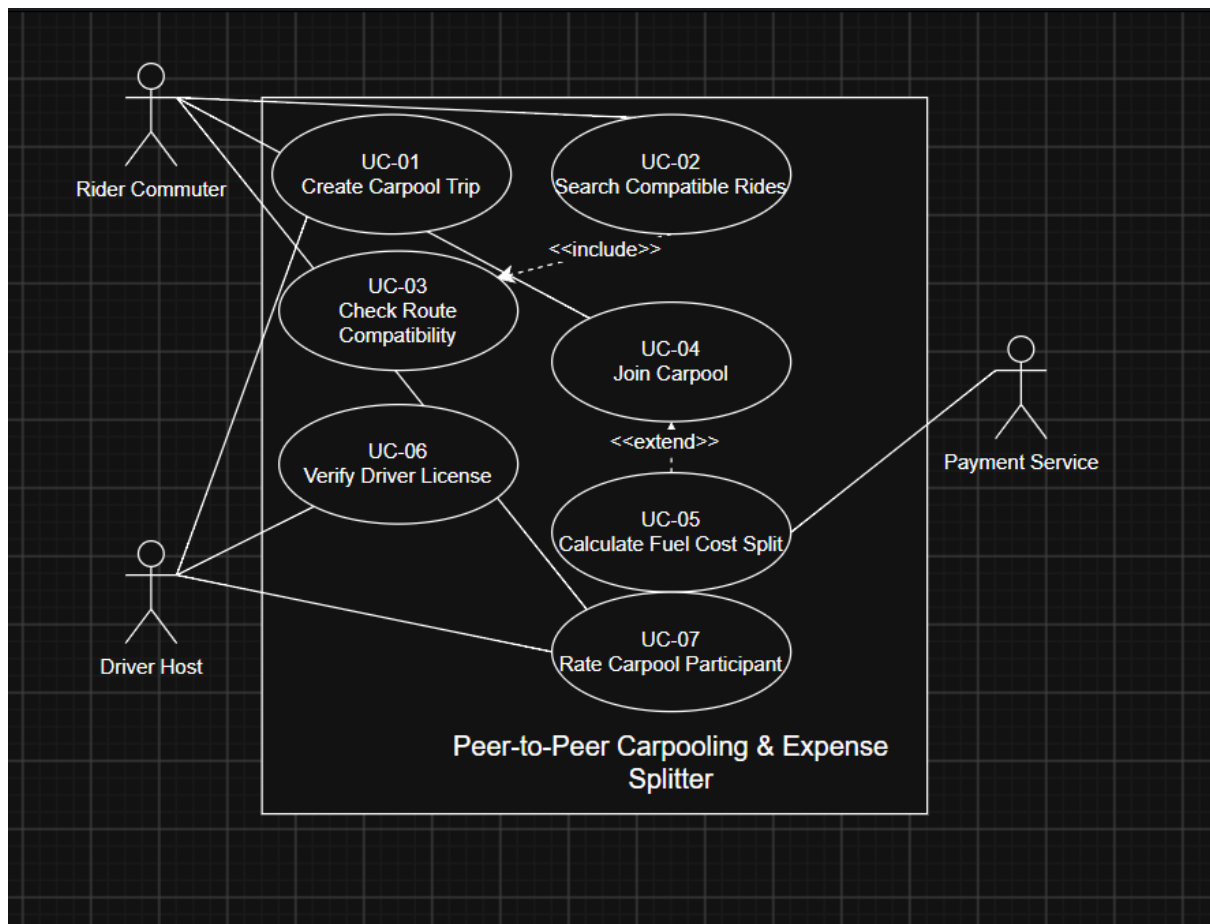
# SOFTWARE ENGINEERING LAB 1

Table

| Req ID | Type       | Description                                                                                                                                            | Priority | Acceptance Criteria                                                                                                                                                                  | Rationale        |
|--------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| FR-001 | Functional | The system shall calculate route compatibility between a driver and passenger and permit matches only when the required detour does not exceed 1.5 km. | High     | Pass: A driver-passenger match with a detour of 1.5 km or less is accepted and the fuel-cost split is calculated. Fail: A match requiring more than 1.5 km detour is accepted.       | Detour tolerance |
| FR-002 | Functional | The system shall allow a Driver Host to create a carpool trip by entering the starting location, destination, travel date, and available seats.        | High     | Pass: A trip is created when all required details are valid and the trip appears in available rides. Fail: The system creates a trip when any required detail is missing or invalid. | Publish routes   |
| FR-003 | Functional | The system shall allow a Rider Commuter to search for available rides based on their starting location, destination, and travel date.                  | High     | Pass: The system displays available rides matching the rider's search and route criteria. Fail: The system displays rides that do not satisfy the specified search criteria.         | Find carpools    |

|         |               |                                                                                                                                                  |        |                                                                                                                                                                                                                      |                         |
|---------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| FR-004  | Functional    | The system shall calculate and display the estimated fuel cost share for each confirmed carpool based on the trip cost and participating riders. | High   | Pass: The system displays a fuel-cost share for every confirmed carpool. Fail: A confirmed carpool is completed without displaying the calculated fuel-cost share.                                                   | Fair expense sharing    |
| FR-005  | Functional    | The system shall allow a Rider Commuter to request and confirm a seat on an available compatible carpool.                                        | High   | Pass: A rider can confirm a seat when the selected carpool has an available seat and the route is compatible. Fail: The system confirms a rider when no seat is available or the route exceeds the detour tolerance. | Participate in carpools |
| NFR-001 | Nonfunctional | The system shall store user ratings, driving license verification documents, and travel histories securely with encrypted backups.               | High   | Pass: Stored sensitive data and backups are encrypted and cannot be accessed by unauthorized users. Fail: Sensitive data or backups are stored without encryption.                                                   | Data protection         |
| NFR-002 | Nonfunctional | The system shall return route compatibility and ride-search results within 3 seconds under normal operating conditions.                          | Medium | Pass: At least 95% of route compatibility and ride-search requests receive results within 3 seconds during testing. Fail: More than 5% of tested requests exceed 3 seconds.                                          | System responsiveness   |

# UML CASE DIAGRAM



## USE CASE FLOW – UC – 04 Join Carpool

Use Case Name

UC-04 – Join Carpool

Primary Actor: Rider Commuter

Supporting Actor: Driver Host

Preconditions

- The rider is logged into the system.
- The rider has entered their starting location and destination.
- There is an available carpool for the selected route and date.
- The selected carpool has at least one available seat.

Postconditions

- The rider is added to the selected carpool.

- The number of available seats is updated.
- The rider receives a confirmation of the booking.
- The estimated fuel cost share is displayed.

#### Main Success Scenario

- The rider searches for available carpools by entering the starting location, destination and travel date.
- The system displays the available compatible rides.
- The rider selects a suitable carpool.
- The system checks whether a seat is available.
- The system checks the route compatibility.
- The rider confirms the request to join the carpool.
- The system adds the rider to the selected carpool.
- The system updates the number of available seats.
- The system calculates the estimated fuel cost share.
- The system displays the booking confirmation.

#### Alternate Flow

##### No Seat Available

- The system finds that the selected carpool is full.
- The rider is not allowed to join the carpool.
- The system displays a message that no seat is available.
- The rider goes back and selects another available ride.

##### Route Not Compatible

- The system finds that the required detour is more than 1.5 km.
- The selected carpool is rejected.
- The system informs the rider that the route is not compatible.
- The rider searches for another compatible ride.

